

Contamination of carcasses with human pathogenic Yersinia enterocolitica 4/O:3 originates from pigs infected on farms.



Pigs are considered as a major reservoir of human pathogenic *Yersinia enterocolitica* and a source of human yersiniosis. However, the transmission route of *Y. enterocolitica* from farm to pork is still unclear. The transmission of pathogenic *Y. enterocolitica* from pigs to carcasses and pluck sets was investigated by collecting samples from 364 individual ear-tagged pigs on the farm and at the slaughterhouse. In addition, isolated strains were analyzed, using pulsed-field gel electrophoresis. Isolation of similar genotypes of pathogenic *Y. enterocolitica* 4/O:3 in animals on the farm and at the slaughterhouse and in carcasses shows that carcass contamination originates from the strains a pig carries during the fattening period. Direct contamination from the carrier pig to its subsequent pluck set is also the primary contamination route for pluck sets, but cross-contamination appears to have a larger impact on pluck set contamination than on carcasses. In this study, the within-farm prevalence of pathogenic *Y. enterocolitica* varied from 0% to 100%, indicating specific farm factors affect the prevalence of *Y. enterocolitica* in pigs. The association of farm factors with the high prevalence of pathogenic *Y. enterocolitica* on farms was studied for the first time, using correlation and two-level logistic regression analyses. Specific farm factors, i.e. drinking from a nipple, absence of coarse feed or bedding for slaughter pigs, and no access of pest animals to pig house, were associated with a high prevalence of pathogenic *Y. enterocolitica* 4/O:3.